

Aparnaa Senthilnathan

315-832-2530 | apar2003@gmail.com | linkedin.com/in/aparnaain | github.com/AparCode | aparcodes.github.io/aparfolio

EDUCATION

Rochester Institute of Technology

Rochester, NY

Bachelor & Master of Science in Computer Science, GPA: 3.36, RIT Presidential Scholar

Aug 2021 - May 2026

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL, HTML, CSS

Full-Stack Engineering: React, FastAPI, REST APIs, Node.js, JSON, API Testing, Object-Oriented Programming

Databases and Tools: MySQL, Relational Database Fundamentals, Data Processing, Data Validation, Git, Docker, Linux, Slurm

AI and Machine Learning: PyTorch, TensorFlow, scikit-learn, OpenCV, Librosa, RAG, ChromaDB, Qdrant

EXPERIENCE

Computer Vision Intern

May 2024 – August 2024

Kitware, Inc.

Carrboro, NC

- Orchestrated distributed data-processing workflows using Slurm and multi-GPU environments, optimizing resource allocation for high-throughput training and evaluation
- Engineered extensible pipeline components for YOLOX workflows, supporting integration and testing of new model architectures
- Automated dataset curation and validation using FiftyOne and PyTorch, standardizing edge-case detection and improving data quality checks

Artificial Intelligence Research Intern

July 2023 - December 2023

Griffiss Institute

Rome, NY

- Designed adversarial stress-testing frameworks to evaluate system reliability, specifically targeting zero-shot classification failures and out-of-distribution data anomalies
- Implemented automated analysis pipelines to quantify system robustness against homonym-based classification failures
- Refactored legacy evaluation scripts into modular components, improving the reproducibility of experiments across ResNet and Vision Transformer model architectures

PROJECTS

MAARA: Multi-Agent Automated Repair Assistant | Academic Team Project

April 2026 - May 2026

- Architected a five-component workflow — Planner, Evaluator, Coder, Critic, and Repair — with clear task boundaries, validation steps, and iterative failure handling
- Built a modular multi-agent software system using Python, Ollama, ChromaDB/Qdrant, and structured feedback loops to improve code-generation quality
- Designed a "Sentinel" strategy supervisor to strictly evaluate pre-generation implementation plans, driving a 13-point pass performance increase over single-agent baselines on the HumanEval benchmark

Mood-y: Voice-Based Mood Music Recommender | Academic Team Project

March 2026 - May 2026

- Developed a full-stack web application using FastAPI and React, exposing REST APIs that mapped user voice inputs to personalized music recommendations
- Implemented backend logic for feature extraction, emotion classification, recommendation ranking, and API response handling using Python, Librosa, scikit-learn, and cosine similarity
- Processed and ranked 81,000+ tracks using weighted scoring and mood-based filtering to generate personalized recommendations

FrequencyPrint: An Automated Deepfake Detection System | Capstone Project

January 2026 - May 2026

- Implemented an AI-driven audio forensics tool in Python and PyTorch, converting audio into spectrogram representations with Librosa for visual pattern analysis
- Trained CNN-based PyTorch models to detect synthetic artifacts in AI-generated covers, achieving up to 61.5% classification accuracy
- Presented research findings and technical methodologies at the Imagine RIT festival and the Graduate Poster Showcase